MSG No. 140.541 $I4/mcm$ [Type I, tetragonal]

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det	TR
1	$\{1\|0\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
2	$\{4^+_{001}\|0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
3	$\{4^-_{001}\|0\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
4	$\{2_{100}\|00\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	1	1
5	$\{2_{010}\|00\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	1	1
6	$\{2_{001}\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	1	1
7	$\{2_{110}\|00\frac{1}{2}\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	1	1
8	$\{2_{1-10}\|00\frac{1}{2}\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	1	1
9	$\{-1\|0\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	1
10	$\{-4^+_{001}\|0\}$	$\begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	1
11	$\{-4^-_{001}\|0\}$	$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}$	−1	1
12	$\{\mathrm{m}_{100}\|00\frac{1}{2}\}$	$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	−1	1
13	$\{\mathrm{m}_{010}\|00\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & \frac{1}{2} \end{bmatrix}$	−1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
14	$\{\mathrm{m}_{001}\|0\}$	$\begin{bmatrix}1&0&0&0\\0&1&0&0\\0&0&-1&0\end{bmatrix}$	-1	1
15	$\{\mathrm{m}_{110}\|00\frac{1}{2}\}$	$\begin{bmatrix}0&-1&0&0\\-1&0&0&0\\0&0&1&\frac{1}{2}\end{bmatrix}$	-1	1
16	$\{\mathrm{m}_{1-10}\|00\frac{1}{2}\}$	$\begin{bmatrix}0&1&0&0\\1&0&0&0\\0&0&1&\frac{1}{2}\end{bmatrix}$	-1	1
17	$\{1\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}1&0&0&\frac{1}{2}\\0&1&0&\frac{1}{2}\\0&0&1&\frac{1}{2}\end{bmatrix}$	1	1
18	$\{4^+_{001}\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}0&-1&0&\frac{1}{2}\\1&0&0&\frac{1}{2}\\0&0&1&\frac{1}{2}\end{bmatrix}$	1	1
19	$\{4^-_{001}\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}0&1&0&\frac{1}{2}\\-1&0&0&\frac{1}{2}\\0&0&1&\frac{1}{2}\end{bmatrix}$	1	1
20	$\{2_{100}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}1&0&0&\frac{1}{2}\\0&-1&0&\frac{1}{2}\\0&0&-1&0\end{bmatrix}$	1	1
21	$\{2_{010}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&1&0&\frac{1}{2}\\0&0&-1&0\end{bmatrix}$	1	1
22	$\{2_{001}\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&-1&0&\frac{1}{2}\\0&0&1&\frac{1}{2}\end{bmatrix}$	1	1
23	$\{2_{110}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}0&1&0&\frac{1}{2}\\1&0&0&\frac{1}{2}\\0&0&-1&0\end{bmatrix}$	1	1
24	$\{2_{1-10}\|\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix}0&-1&0&\frac{1}{2}\\-1&0&0&\frac{1}{2}\\0&0&-1&0\end{bmatrix}$	1	1
25	$\{-1\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}-1&0&0&\frac{1}{2}\\0&-1&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	-1	1
26	$\{-4^+_{001}\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}0&1&0&\frac{1}{2}\\-1&0&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	-1	1
27	$\{-4^-_{001}\|\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix}0&-1&0&\frac{1}{2}\\1&0&0&\frac{1}{2}\\0&0&-1&\frac{1}{2}\end{bmatrix}$	-1	1

continued ...

Table 1

No.	tag	matrix (polar)	det	TR
28	$\{\mathrm{m}_{100}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} -1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	-1	1
29	$\{\mathrm{m}_{010}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & -1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	-1	1
30	$\{\mathrm{m}_{001}\vert\frac{1}{2}\frac{1}{2}\frac{1}{2}\}$	$\begin{bmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & -1 & \frac{1}{2} \end{bmatrix}$	-1	1
31	$\{\mathrm{m}_{110}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 0 & -1 & 0 & \frac{1}{2} \\ -1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	-1	1
32	$\{\mathrm{m}_{1-10}\vert\frac{1}{2}\frac{1}{2}0\}$	$\begin{bmatrix} 0 & 1 & 0 & \frac{1}{2} \\ 1 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & 0 \end{bmatrix}$	-1	1